

Oxidation of organic compounds

Learning outcomes

By the end of this section you should be able to deduce equations to show changes in the functional groups during the oxidation of primary and secondary alcohols, including the two-step reaction in the oxidation of primary alcohol.

The environmental impact of plastics is an increasing concern and a threat to ecosystems around the globe. Scientists are studying the creation of products that could replace plastics and also ways to reduce the plastic that we already have introduced into the environment.

Carboxylic acids can be used to make biodegradable polymers, such as polylactic acid (PLA), which is a biodegradable polymer made from lactic acid, a carboxylic acid derived from renewable sources such as corn starch. Ketones and carboxylic acids can also be used as solvents or reactants in the degradation of pollutants. For instance, carboxylic acids can be used to extract heavy metals from contaminated soils, while ketones can be used to break down organic pollutants, such as polycyclic aromatic hydrocarbons (PAHs) or bisphenol A (BPA), which are commonly found in industrial and household waste.

How can we have access to more of these useful ketones and carboxylic acids?

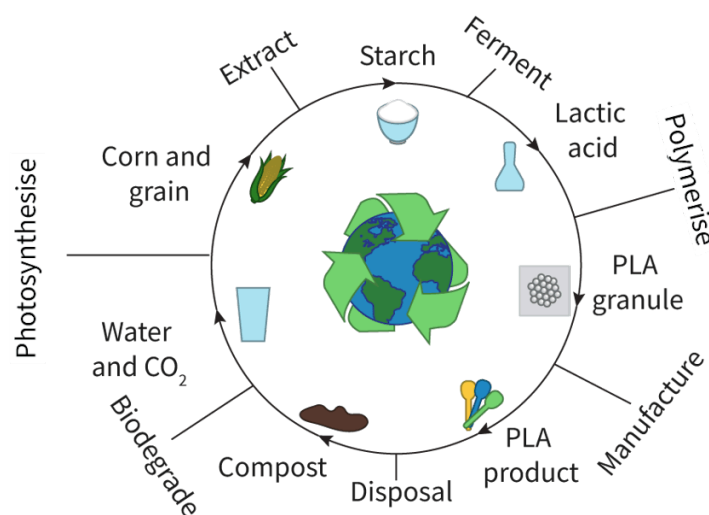


Figure 1. PLA production and use.

 More information for figure 1

The circular diagram illustrates the production and lifecycle of Poly(lactic acid) (PLA). At the center is an image of the Earth surrounded by a recycling symbol, indicating the eco-friendly nature of the process. The cycle begins with 'Photosynthesis', where corn and grain are grown, labeled as 'Corn and grain'. From there, 'Extract' leads to 'Starch' and then 'Ferment', producing 'Lactic acid'. The process continues with 'Polymerise' to form 'PLA granule', which is then 'Manufacture'd into 'PLA product'. 'PLA product' can be 'Disposal'd or 'Biodegrade'd into 'Compost'. 'Compost' is used for 'Photosynthesis' of 'Corn and grain'.

where 'PLA granule' is formed. 'Manufacture' creates 'PLA product'. After use, the product is 'Disposed' and goes through 'Compost'. Finally, it 'Biodegrades' to produce 'Water and CO₂', which then returns to 'Photosynthesis', completing the cycle.

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Alcohol structure

You might recall that the alcohols (see [subtopic S3.2](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>)) have a hydroxyl (–OH) group, the ketones have a carbonyl (=O) group and the carboxylic acids have a carboxyl (–COOH) group.

Table 1. Alcohols, carboxylic acids and ketones.

C o m p o u n d		
	Functional group	Example
Al co ho l	$\text{R} - \text{OH}$ Hydroxyl	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{O}$  More information The image depicts the chemical structure formula of Butanol. It is shown as a molecule chain: CH₃CH₂CH₂CH₂OH. The letters represent the atoms in the molecule: C for carbon, H for hydrogen, and OH for the hydroxyl group. The chain of letters is displayed horizontally, indicating the linear structure of the Butanol molecule. [Generated by AI] Butanol
Ca rb		

C o m p o u n d	Functional group	Example
ox y l i c a c i d	More information The image is a diagram showing the chemical structure of a carboxyl group. It consists of a carbon (C) atom double-bonded to an oxygen (O) atom, and single-bonded to a hydroxyl group (OH). Additionally, the carbon atom is single-bonded to a variable group represented as R. This structural formula represents the functional group typical in carboxylic acids. [Generated by AI] Carboxyl	CH_3COOH Ethanoic acid
Ke to ne	More information The image is a chemical structure diagram of a ketone molecule. It shows a carbon (C) atom double-bonded to an oxygen (O), labeled as "O=C". Attached to the carbon are two groups, indicated as "R" and "R'", which represent the rest of the molecule and can vary. These groups form single bonds with the carbon atom to complete the angular structure typical of simple ketones. [Generated by AI]	CH_3COCH_3 Propanone

C o m p o u n d	Functional group	Example
	Ketone	

Earlier in your studies you learned that alcohols can be classified as primary, secondary or tertiary (see section S3.2.6 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structural-isomerism-id-45665/>)), depending on the number of carbon atoms bonded to the carbon atom that is directly bonded to the functional group. **Figure 2** shows some examples of primary, secondary and tertiary alcohols.

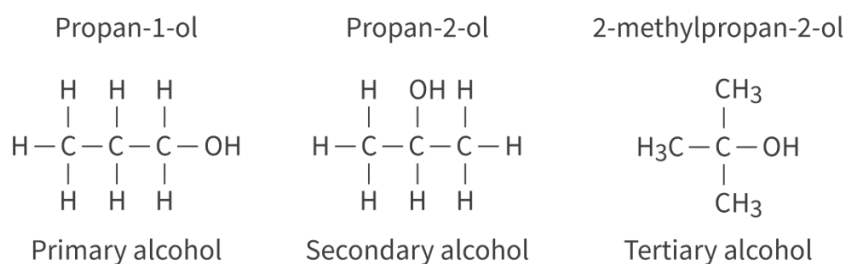


Figure 2. Some examples of primary, secondary and tertiary alcohols.

 More information for figure 2

The image depicts molecular structures of three types of alcohols, highlighting their classification as primary, secondary, and tertiary. The first structure is for Propan-1-ol, a primary alcohol, shown with three carbon atoms in a chain. Each carbon is bonded with hydrogen atoms, and the functional group (-OH) is attached to the end carbon. The second structure is for Propan-2-ol, a secondary alcohol, with the -OH group attached to the middle carbon of the three-carbon chain. Hydrogen atoms are bonded to the other available sites on the carbon atoms. The third structure is for 2-methylpropan-2-ol, a tertiary alcohol, where the -OH group is attached to a central carbon atom that is also bonded to a methyl group (CH₃) in addition to other carbon atoms. This diagram visually differentiates the alcohols by illustrating their different functional group attachments to the carbon chain.

[Generated by AI]

Check your understanding of the classification of alcohols by answering these questions.



Exercises 1 and 2

Click a question to answer



1. What type of alcohol is 2-methylpropan-2-ol?



2. Butan-2-ol is classified as a _____ alcohol.

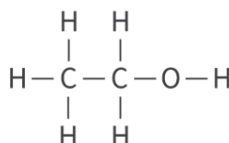
✓ Check answers

You also learned previously that oxidation can be defined as the addition of oxygen or the removal of hydrogen (see [section R3.2.1–2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/>)).

Figure 3. A comparison of oxidation and reduction.

Now, let's compare ethanol, ethanal and ethanoic acid in terms of their oxidation.

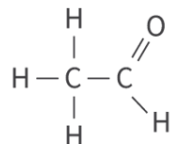
Which one is the most oxidised or reduced?



More information

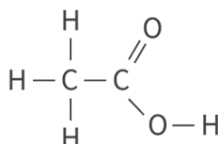
The image shows the structural formula of an ethanol molecule. It is represented by a linear arrangement of carbon (C), hydrogen (H), and oxygen (O) atoms connected by single bonds. The formula follows this sequence: a central carbon atom (C) is bonded to two hydrogen atoms (H) on the left and right sides. The carbon atom is also bonded upwards to another carbon atom, which is connected to two hydrogen atoms above and below. This second carbon atom is attached to an oxygen atom (O) on its right, which in turn is bonded to a hydrogen atom. This sequence forms the molecular configuration CH₃CH₂OH, representing ethanol.

[Generated by AI]

[More information](#)

The image is a chemical structure diagram representing the acetaldehyde molecule. It consists of two carbon atoms bonded together, labeled as 'C'. One of the carbon atoms is single-bonded to three hydrogen atoms labeled 'H'. The other carbon atom is double-bonded to an oxygen atom labeled 'O' and single-bonded to a hydrogen atom. This structure shows the typical arrangement of atoms in an acetaldehyde molecule, with the oxygen atom forming a double bond with one of the carbon atoms to create the carbonyl group.

[Generated by AI]

[More information](#)

The image shows the chemical structure of ethanol, a primary alcohol. It is depicted with a central carbon (C) atom bonded to three hydrogen (H) atoms and one hydroxyl (OH) group. The carbon is also bonded to another carbon which is double-bonded to an oxygen (O) atom. The hydroxyl group is bonded to the second carbon atom. This structure is a typical representation of ethanol, commonly found in alcoholic beverages. The schematic highlights the arrangement of atoms, demonstrating typical bond angles and orientations associated with organic chemical structures.

[Generated by AI]

Ethanol
(primary alcohol)

Ethanal
(aldehyde)

Ethanoic acid
(carboxylic acid)

Figure 4. Ethanol, ethanal and ethanoic acid.

You might be able to identify that the aldehyde has two fewer hydrogens than the alcohol, which means it is more oxidised than the alcohol. It might be also clear than in comparison, the carboxylic acid has one more oxygen than the aldehyde, meaning it is more oxidised.

Oxidation of primary alcohols

Primary alcohols can be oxidised into aldehydes and carboxylic acids when in the presence of an oxidising agent [O] according to the reactions in **Figure 5**. Notice that the oxidation number of the carbon in the alcohol will increase and either an aldehyde or a carboxylic acid will be produced.

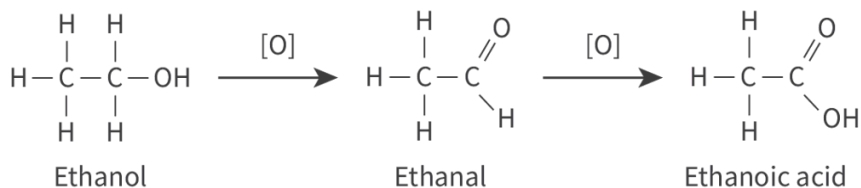


Figure 5. The oxidation of alcohols into aldehydes and carboxylic acids.

 More information for figure 5

The image is a chemical diagram illustrating the oxidation process of ethanol into ethanal and further into ethanoic acid. The structures are depicted as follows:

1. Ethanol (C₂H₅OH): Two carbon atoms are shown, with the first carbon bonded to three hydrogen atoms and the second carbon bonded to one hydrogen and a hydroxyl group (OH).
2. An arrow marked with "[O]" indicates the presence of an oxidizing agent that facilitates the reaction to form ethanal.
3. Ethanal (C₂H₄O): The first carbon remains bonded to three hydrogen atoms. The second carbon now forms a double bond with an oxygen atom and is also bonded to one hydrogen atom, indicating the formation of an aldehyde.
4. Another arrow marked with "[O]" shows further oxidation to ethanoic acid.
5. Ethanoic acid (C₂H₄O₂): The structure has the first carbon bonded to three hydrogen atoms. The second carbon is bonded to a hydroxyl group (OH) and double-bonded to an oxygen atom, representing the carboxylic acid formation.

[Generated by AI]

For example, glycerol (C₃H₈O₃) is a naturally occurring alcohol that can be used as a sweetener for low-fat food products, and it can be oxidised into glyceraldehyde or glyceric acid in the presence of an oxidising agent (**Figure 6**).

In the human body, glycerol is involved in the balance of cellular redox, crucial for the proper cellular functioning and the prevention of oxidative stress that can damage cellular components.

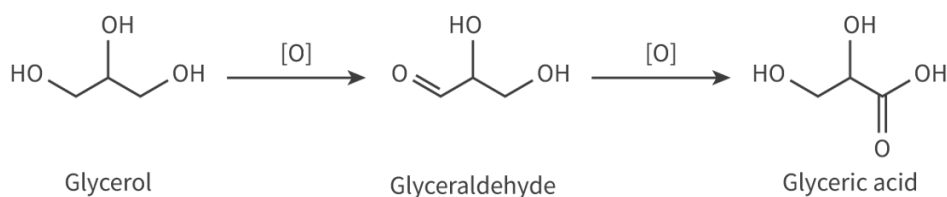


Figure 6. The oxidation of glycerol to glyceraldehyde and glyceric acid.

 More information for figure 6

The image is a chemical reaction diagram illustrating the oxidation process of glycerol to form glyceraldehyde and subsequently glyceric acid. The diagram is composed of three main chemical structures connected by arrows indicating the flow of reaction. On the left, the structure of glycerol is depicted, which consists of a three-carbon chain with each carbon (C) attached to a hydroxyl group (OH). An arrow labeled with [O] points to the next structure, glyceraldehyde, a three-carbon aldehyde with a double-bonded oxygen on the first carbon. A similar arrow labeled with [O] leads to glyceric acid, which has a similar three-carbon structure but features both a double-bonded oxygen and an additional hydroxyl group.

[Generated by AI]

The presence of the hydroxyl functional group makes alcohols susceptible to oxidation in the presence of a suitable oxidising agent. You do not need to know the names of suitable oxidising agents, but an understanding of their function is important. During the reaction, the oxidising agent removes electrons from the alcohol (or aldehyde) and becomes reduced. In the process, the oxidising agent changes colour, as shown in **Figure 7** and **Figure 8**.

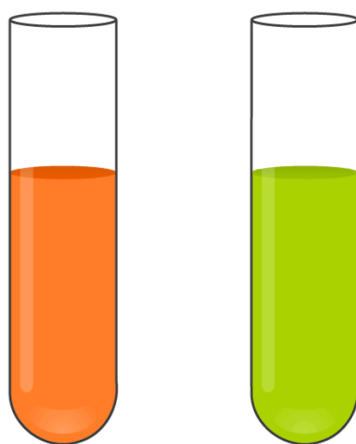


Figure 7. The colour change from orange to green when an alcohol undergoes oxidation with acidified potassium dichromate(VI).

Making connections

Why is there a colour change when an alcohol is oxidised by a transition element compound? (see [subtopic S3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/)).

Transition element compounds have different colours depending on the oxidation state of the transition element. For example, the dichromate ion ($\text{Cr}_2\text{O}_7^{2-}$) with Cr^{6+} ions forms an orange solution which changes to a green colour as it is reduced to Cr^{3+} (Figure 7).

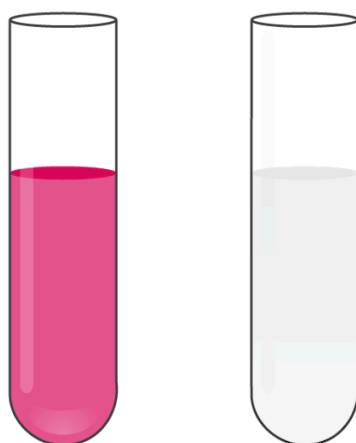


Figure 8. The colour changes from purple to colourless when an alcohol undergoes oxidation with acidified potassium manganate(VII) solution.

Study skills

Names and formulas of specific oxidising agents and the mechanism will not be assessed.

Primary alcohols, such as ethanol, produce aldehydes or carboxylic acids, depending on the degree of oxidation (partial or complete). The partial oxidation of a primary alcohol produces an aldehyde. The reaction is carried out with an excess of alcohol as in **Figure 9**. Note the use of [O] to represent the oxidising agent in the reaction.

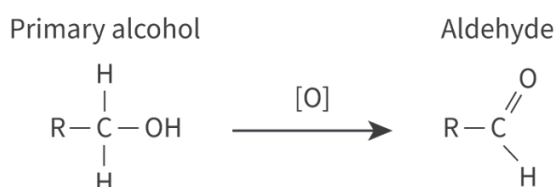


Figure 9. The partial oxidation of a primary alcohol to an aldehyde.

 More information for figure 9

The image is a diagram illustrating the partial oxidation of a primary alcohol to an aldehyde. On the left, there is a structural representation of a primary alcohol with the generic formula $\text{R}-\text{C}-\text{OH}$, where the carbon (C) is bonded to two hydrogen atoms (H) and a hydroxyl group (OH). To the right of this structure, an arrow points towards another structure labeled 'Aldehyde'. Above the arrow, the symbol [O] is used to represent the oxidizing agent involved in the reaction. The structure on the right represents an aldehyde with the formula $\text{R}-\text{C}-\text{H}$, where the carbon is double bonded to an oxygen atom (O) and single bonded to a hydrogen atom (H). This diagram communicates the chemical transformation from a primary alcohol to an aldehyde through partial oxidation.

[Generated by AI]

Figure 10 shows the partial oxidation of the primary alcohol ethanol to the aldehyde ethanal.

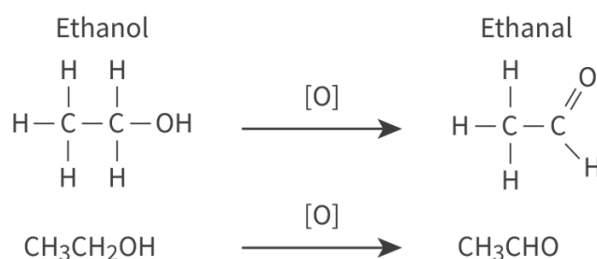


Figure 10. The partial oxidation of ethanol to ethanal.

 More information for figure 10

The image depicts a chemical diagram that illustrates the partial oxidation of ethanol to ethanal. On the left, the ethanol molecule is shown with the chemical formula $\text{C}_2\text{H}_5\text{OH}$, consisting of a hydroxyl group (OH) bonded to a 2-carbon chain, with each carbon atom bonded to hydrogen atoms (H). To the right, after the oxidation process indicated by an arrow and [O] placeholder, the ethanol molecule is transformed into ethanal. The ethanal molecule is represented by the chemical formula CH_3CHO and includes an aldehyde group (CHO) attached to the carbon chain, indicating the replacement of one hydrogen atom with an oxygen atom double-bonded to a carbon atom.

[Generated by AI]

Distillation apparatus

If the aldehyde is the desired product, it can be removed as it forms by distillation to prevent further oxidation (**Figure 11**). This is possible because the boiling point of the aldehyde is lower than that of both the alcohol and the carboxylic acid. As the aldehyde evaporates and rises up the distillation column, it passes through the condenser where it condenses back to a liquid and runs down into the flask.

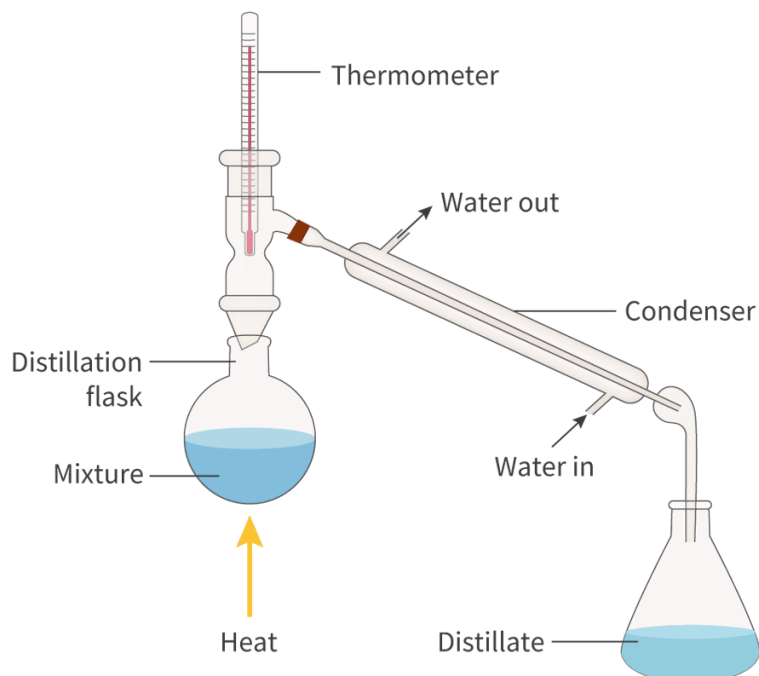


Figure 11. Distillation apparatus used to separate components of a mixture with different boiling points.

 More information for figure11

The image is a labeled diagram of a distillation apparatus used for separating components of a mixture based on their boiling points. The apparatus consists of several key components:

1. A distillation flask containing the mixture, which is heated.
2. A thermometer is inserted into the apparatus to measure the temperature.
3. The vapor rises and moves through the condenser.
4. The condenser has labels for 'Water in' at the lower end and 'Water out' at the upper end to control the condensation process.
5. The vapor condenses back into liquid in the condenser tube.
6. The liquid, known as distillate, is collected in a receiving flask.

The diagram shows the process of heating the mixture to facilitate evaporation of the component with a lower boiling point, which then condenses and is collected separately as a distillate.

[Generated by AI]

Making connections

How does the nature of the functional group in a molecule affect its physical properties, such as boiling point? (see [subtopic S3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>)).

The nature of the functional group in a molecule can significantly affect its physical properties, including the boiling point, due to the intermolecular forces present between the particles of the substance.

Boiling point is a measure of the temperature at which a liquid changes state from liquid to gas. The stronger the intermolecular forces between the molecules, the higher the boiling point of the substance.

In general, we can say that alkanes have lower boiling points as they can only form London forces between their molecules. Alcohols contain a hydroxyl (—OH) functional group, which can participate in hydrogen bonding with other molecules, therefore alcohols generally have higher boiling points than similar-sized hydrocarbons. Aldehydes and ketones contain a carbonyl (C=O) functional group, which can participate in dipole-dipole interactions which are weaker than hydrogen bonding, so aldehydes and ketones generally have lower boiling points than alcohols of similar size. Finally, carboxylic acids contain a carboxyl (—COOH) functional group, which can participate in hydrogen bonding and dipole-dipole with other molecules, which generally gives them the highest boiling points.

Reflux apparatus

The complete oxidation of a primary alcohol produces a carboxylic acid as shown in **Figure 12**. The conditions used are heat under reflux with an excess of the oxidising agent.

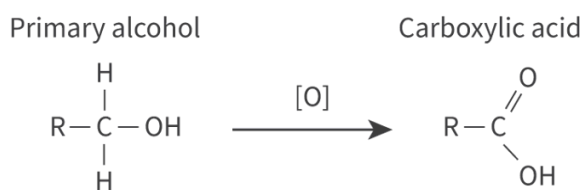


Figure 12. The complete oxidation of a primary alcohol produces a carboxylic acid.

 More information for figure 12

The image is a chemical reaction diagram illustrating the oxidation process of a primary alcohol to form a carboxylic acid. The diagram shows a structural formula of a primary alcohol on the left side, labeled as 'Primary alcohol,' with a generic structure: $\text{R-CH}_2\text{OH}$. An arrow labeled with "[O]" points towards the right side, indicating the addition of an oxidizing agent. On the right, the resultant carboxylic acid is represented, labeled as 'Carboxylic acid,' with the structure: R-COOH . The arrow signifies the conversion process facilitated by oxidation.

[Generated by AI]

The complete oxidation of ethanol to ethanoic acid is shown in **Figure 13**.

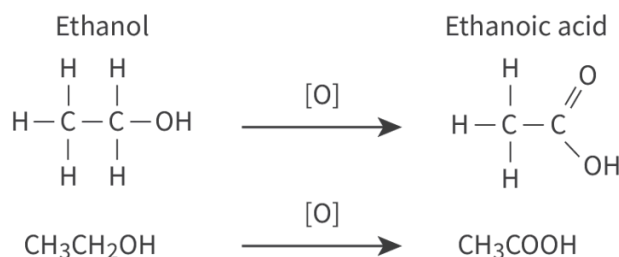


Figure 13. The complete oxidation of ethanol to produce ethanoic acid.

 More information for figure 13

The image displays a chemical reaction showing the complete oxidation of ethanol to ethanoic acid. The structural formula for ethanol is shown on the left with the atoms and bonds depicted as: two carbon atoms with three hydrogen atoms bonded to the first carbon and one hydroxyl group (OH) bonded to the second carbon. Below the structural formula is the molecular formula 'CH₃CH₂OH.' An arrow points to the right, indicating the reaction process with an '[O]' above and below the arrow, representing oxidation. On the right, the structural formula for ethanoic acid is depicted with a carboxyl group (COOH) bonded to a carbon atom, which is also connected to two other hydrogen atoms. Below the structural formula, the molecular formula 'CH₃COOH' is shown. This visual representation illustrates the conversion of ethanol to ethanoic acid through an oxidation process.

[Generated by AI]

If the carboxylic acid is the required product, the mixture containing the alcohol and oxidising agent is heated under reflux (**Figure 14**). As the mixture is heated, the vapours rise up the condenser, cool down and condense back into a liquid. This allows the alcohol and the oxidising agent to remain in contact for a prolonged period of time and results in complete oxidation.

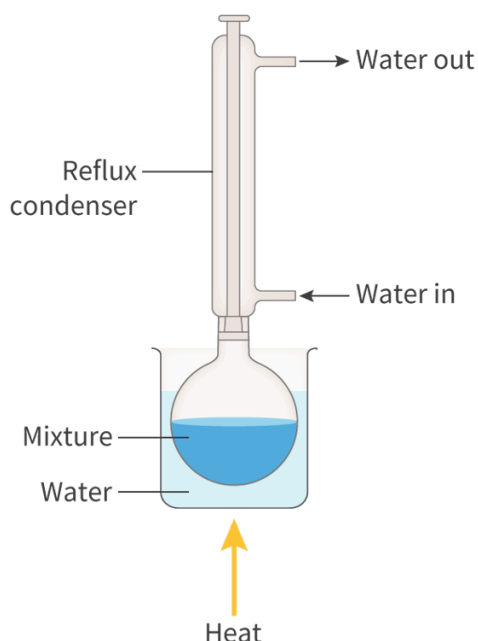


Figure 14. Apparatus for heating under reflux. More information for figure 14

The image shows a diagram of an apparatus for heating under reflux. It consists of a round-bottom flask containing a blue liquid labeled as the mixture. The flask sits in a larger container filled with water. Underneath this container is an arrow pointing upward labeled "Heat," indicating the application of heat. Above the flask is a vertical tube labeled "Reflux condenser" with arrows and labels showing "Water in" on one side and "Water out" on the other. This setup allows vapors to rise, condense in the condenser, and return to the flask, maintaining the reaction mixture in a closed system.

[Generated by AI]

Aldehyde has a lower boiling point than carboxylic acid due to the differences in the intermolecular forces:

- Aldehyde – weaker dipole–dipole intermolecular forces
- Carboxylic acid – stronger hydrogen bonds intermolecular forces

Worked example 1

Draw and name the product in the oxidation of butan-1-ol under reflux.

Under reflux, the primary alcohol will oxidise completely into a carboxylic acid, in this case butanoic acid.

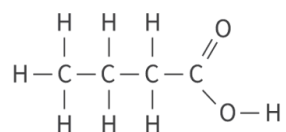


Figure 15. Butanoic acid.

**Worked example 2**

Draw and name the product of the oxidation of hexan-1-ol under distillation.

Under distillation, the primary alcohol will oxidise partially into an aldehyde, in this case hexanal.

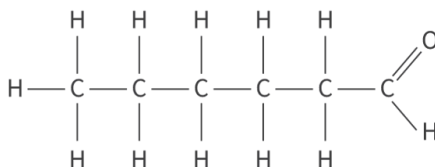


Figure 16. The full structural formula of hexanal.



Oxidation of secondary alcohols

The products of the oxidation reaction depend on whether the alcohol is primary or secondary, and on the reaction conditions (note that tertiary alcohols do not undergo oxidation).

What would oxidation look like for a secondary alcohol? Where will hydrogen atoms be removed or oxygen atoms added? How far can the oxidation occur?

The oxidation of a secondary alcohol produces only one product – a ketone as in **Figure 17**. The required reaction conditions are heat under reflux with a suitable oxidising agent such as acidified potassium dichromate(VI).

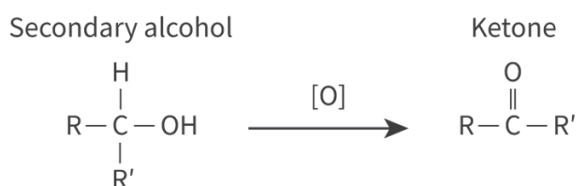


Figure 17. Secondary alcohols are oxidised to form ketones.

More information for figure 17

The image shows a chemical reaction equation where a secondary alcohol is oxidized to form a ketone. On the left side of the equation, there's a structural formula for secondary alcohol labeled "Secondary alcohol," with the structure: $\text{R}_1-\text{C}(-\text{OH})(-\text{H})-\text{R}_2$. An arrow labeled "[O]" points from the secondary alcohol to the ketone on the right side. The ketone is labeled "Ketone," with the structure: $\text{R}_1-\text{C}(=\text{O})-\text{R}_2$. This illustration depicts the oxidation process where the hydrogen bonded to the carbon in the alcohol is replaced by an oxygen to form the carbonyl group of a ketone.

[Generated by AI]

Figure 18 shows the oxidation of the secondary alcohol, propan-2-ol, to produce propanone.

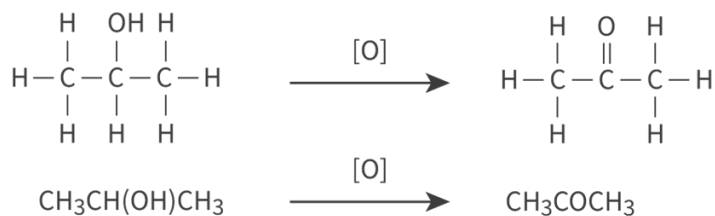


Figure 18. The oxidation of propan-2-ol to propanone.

 More information for figure 18

The image is a chemical diagram illustrating the oxidation process of propan-2-ol to propanone. On the left side, the structure of propan-2-ol is shown, represented by a chain of three carbon atoms. The first carbon atom is bonded to three hydrogen atoms, the second carbon is bonded to one hydrogen atom and one hydroxyl group (OH), and the third carbon is bonded to three hydrogen atoms. Below the structure is the molecular formula $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$. An arrow labeled with [O] points from the propan-2-ol structure to the right side, indicating an oxidation reaction. On the right side, the structure of propanone is shown, represented by a similar chain of three carbon atoms. The first carbon atom is bonded to three hydrogen atoms, the second carbon is double-bonded to an oxygen atom and bonded to another carbon atom, and the third carbon is bonded to three hydrogen atoms. Below this structure is the molecular formula CH_3COCH_3 . The diagram effectively shows the transformation of the secondary alcohol to a ketone through the oxidation process.

[Generated by AI]

Worked example 3

Draw and name the product of the oxidation of butan-2-ol under reflux.

Under reflux, the secondary alcohol will oxidise completely into a ketone, in this case butanone.

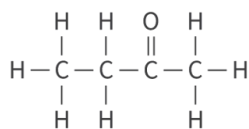


Figure 19. Butanone.



Making connections

What is the difference between the combustion and oxidation of an alcohol? (see subtopic R1.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43475/>)).

Alcohols combust in the presence of oxygen and produce carbon dioxide and water, as well as energy in the form of heat and light. The combustion of an alcohol is a redox reaction in which the alcohol is oxidised and the oxygen is reduced.

Alcohols can also be oxidised through the addition of an oxidising agent. This process can be used to convert alcohols to other functional groups, such as aldehydes or carboxylic acids.

The key difference between the combustion and oxidation of alcohols is the reaction rate and the nature of the products. Combustion is a rapid process that typically results in the complete oxidation of the alcohol to carbon dioxide and water, while the oxidation of alcohols is a slower process that can be controlled to produce specific products, such as aldehydes or carboxylic acids, depending on the choice of oxidising agent and reaction conditions, like a distillation or reflux apparatus.

Activity

- **IB learner profile attribute:**
 - Inquirer
 - Thinker
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual/pair

Real-world applications of the oxidation of alcohols

Research how primary alcohols are oxidised to aldehydes and carboxylic acids, and how these reactions are relevant to biological and industrial processes.

- Compare the oxidation of different types of alcohols (primary, secondary and tertiary) using acidified sodium dichromate as an oxidising agent.
- Explore real-world applications of alcohol oxidation, such as the biological metabolism of ethanol and methanol, the reflux and distillation setups, the commercial production of various types of vinegar or kombucha using acetic acid bacteria, or the production of acetic acid from ethanol.
- You might consider investigating the common uses and commercial production of ethanoic acid (acetic acid) or methanoic acid (formic acid) — two very important compounds.

Share your report with a partner and reflect on what you have learned.